

Helical Spring Vibrations

Problem Statement

Practice Ex. 5.14: The stiffness of a helical spring is such that it stretches 1 mm for every 1.5 N of load. Find the number of vibrations it will make per minute if it is disturbed from rest when carrying a mass of 6.5 kg.

Solution

Step 1: Determine the Spring Constant

The spring constant k is given by:

$$k = \frac{F}{x}$$

Given that the spring stretches by 1 mm (0.001 m) for every 1.5 N of force:

$$k = \frac{1.5}{0.001} = 1500 \text{ N/m}$$

Step 2: Natural Frequency of Vibration

The natural frequency of a mass-spring system is given by:

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

where:

- $k = 1500 \text{ N/m}$ (spring constant),

- $m = 6.5 \text{ kg}$ (mass).

Substituting the values:

$$f = \frac{1}{2\pi} \sqrt{\frac{1500}{6.5}}$$

Calculating inside the square root:

$$\frac{1500}{6.5} \approx 230.77$$

Taking the square root:

$$\sqrt{230.77} \approx 15.19$$

Thus:

$$f \approx \frac{15.19}{2\pi}$$

Approximating:

$$f \approx \frac{15.19}{6.283} \approx 2.42 \text{ Hz}$$

Step 3: Vibrations Per Minute

Since $1 \text{ Hz} = 1 \text{ cycle per second}$, the number of vibrations per minute is:

$$N = 60 \times f$$

$$N = 60 \times 2.42 \approx 145.2$$

Thus, the number of vibrations per minute is **145** (rounded to the nearest whole number).